

MATERIAL SAFETY DATA SHEET

1. PRODUCT AND MANUFACTURER IDENTIFICATION

LAMBDA-CYHALOTHRIN 10% W/V CS

SUPPLIER NAME:	Shandong Huimin Vanda Biological Technology Co., Ltd.
SUPPLIER ADDRESS:	Jinan Plant Protection Technology Industrial Park, Hanmiao Town, Shanghe County, Jinan City, Shandong Province, China
SUPPLIER PHONE:	+86 531-84981777
SUPPLIER FAX:	+86 543-2232777
CHEMICAL NAME	(R)-a-cyano-3-phenoxybenzyl(1S)-cis-3-[(Z)-2-chloro-3,3,3-trifluoropropenyl]-2,2-dimethylcyclopropanecarboxylate (S)-a-cyano-3-phenoxybenzyl(1R)-cis-3-[(Z)-2-chloro-3,3,3-trifluoropropenyl]-2,2-dimethylcyclopropanecarboxylate
CAS NUMBER	91465-08-6
MOLECULAR FORMULA	C ₂₃ H ₁₉ ClF ₃ NO ₃
MOLECULAR WEIGHT	449.85
CHEMICAL FAMILY	Pyrethroid

2. HAZARDS IDENTIFICATION

CLASSIFICATION OF THE SUBSTANCE OR MIXTURE

Acute toxicity, Oral (Category 4), H302

Acute toxicity, Inhalation (Category 4), H332

Chronic aquatic toxicity (Category 1), H410



PICTOGRAMS

Signal Words Warning

Hazard statements

H302 Harmful if swallowed.

H332 Harmful if inhaled.

H410 Very toxic to aquatic life with long lasting effects.

Precautionary statement(s)

P261: Avoid breathe dust/fume/gas/mist/vapours/spray.

P264: Wash thoroughly after handling.

P270: Do not eat, drink or smoke when using this product.

P271: Use only outdoors or in a well-ventilated area.

P273: Avoid release to the environment.

P312: Call a POISON CENTER or doctor/physician if you feel unwell.

P330: Rinse mouth.

P391: Collect spillage.

P301 + P312: IF SWALLOWED: call a POISON CENTER or doctor/physician.

P304 + P340: IF INHALED: Remove victim to fresh air and keep at rest in a position comfortable for breathing.

P501: Dispose of contents/container in accordance with local regulations.

3. COMPOSITION / INFORMATION ON INGREDIENTS

Lambda-cyhalothrin	10% W/V
Other ingredients	up to 1L

4. FIRST - AID MEASURES

DESCRIPTION OF FIRST AID MEASURES

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GENERAL ADVICE:	Consult a physician. Show this safety data sheet to the doctor in attendance.
INHALATION:	If breathed in, move person into fresh air. If not breathing, give artificial respiration. Consult a physician.
SKIN CONTACT:	Wash off with soap and plenty of water. Consult a physician.
EYE CONTACT:	Flush eyes with water as a precaution.
INGESTION:	Never give anything by mouth to an unconscious person. Rinse mouth with water. Consult a physician.
MOST IMPORTANT SYMPTOMS AND EFFECTS, BOTH ACUTE AND DELAYED:	The most important known symptoms and effects are described in the labelling
INDICATION OF ANY IMMEDIATE MEDICAL ATTENTION AND SPECIAL TREATMENT NEEDED:	No data available

5. FIRE FIGHTING MEASURES

EXTINGUISHING MEDIA:	Suitable extinguishing media: water spray, carbon dioxide, foam, dry powder Unsuitable extinguishing media for safety reasons: water jet
SPECIAL HAZARDS ARISING FROM THE SUBSTANCE OR MIXTURE	carbon monoxide, Carbon dioxide, Hydrogen chloride, nitrogen oxides, organochloric compounds The substances/groups of substances mentioned can be released in case of fire.
ADVICE FOR FIRE-FIGHTERS	Special protective equipment: Wear self-contained breathing apparatus and chemical-protective clothing.

6. ACCIDENTAL RELEASE MEASURES

PERSONAL PRECAUTIONS, PROTECTIVE EQUIPMENT AND EMERGENCY PROCEDURES

Wear respiratory protection. Avoid dust formation. Avoid breathing vapours, mist or gas. Ensure adequate ventilation. Evacuate personnel to safe areas. Avoid breathing dust.

ENVIRONMENTAL PRECAUTIONS

Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.

METHODS AND MATERIALS FOR CONTAINMENT AND CLEANING UP

Pick up and arrange disposal without creating dust. Sweep up and shovel. Keep in suitable, closed containers for disposal.

7. HANDLING AND STORAGE

PRECAUTIONS FOR SAFE HANDLING

Avoid contact with skin and eyes. Avoid formation of dust and aerosols.

Provide appropriate exhaust ventilation at places where dust is formed.

CONDITIONS FOR SAFE STORAGE, INCLUDING ANY INCOMPATIBILITIES

Store in cool place. Keep container tightly closed in a dry and well-ventilated place.

Storage class (TRGS 510): Non-combustible, acute toxic Cat. 1 and 2 / very toxic hazardous materials

8. EXPOSURE CONTROLS / PERSONAL PROTECTION

EYE PROTECTION:	Safety glasses with side-shields. Use equipment for eye protection tested and approved under appropriate government standards
RESPIRATORY PROTECTION:	Where risk assessment shows air-purifying respirators are appropriate use respirator cartridges as a backup to engineering controls. If the full-face supplied air respirator. Use respirators and components tested and approved under appropriate government standards

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SKIN PROTECTION	Handle with gloves. Gloves must be inspected prior to use. Use proper glove removal technique (without touching glove's outer surface) to avoid skin contact with this product. Dispose of contaminated gloves after use in accordance with applicable laws. Wash and dry hands.
BODY PROTECTION	Complete suit protecting against chemicals, The type of protective equipment must be selected according to the concentration and amount of the dangerous substance at the specific workplace.
CONTROL OF ENVIRONMENTAL EXPOSURE	Prevent further leakage or spillage if safe to do so. Do not let product enter drains. Discharge into the environment must be avoided.
ENGINEERING CONTROLS	Handle in accordance with good industrial hygiene and safety practice. Wash hands before breaks and at the end of workday

9. PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE FORM:	White to off-White Liquid
ODOUR	No data available
ODOUR THRESHOLD	No data available
PH	5.43
MELTING POINT/FREEZING POINT	No data available
INITIAL BOILING POINT AND BOILING RANGE	No boiling point before decomposition
FLASH POINT	No data available
EVAPORATION RATE	No data available
FLAMMABILITY (SOLID, GAS)	not considered highly flammable
UPPER EXPLOSION LIMIT:	No Explosive
LOWER EXPLOSION LIMIT:	No Explosive
VAPOUR PRESSURE	No data available
VAPOUR DENSITY	No data available
RELATIVE DENSITY	1.045g/cm ³
WATER SOLUBILITY	4 µg/L at pH 5.0
PARTITION COEFFICIENT: NOCTANOL/WATER	log Pow at 20°C : 5.5
SURFACE TENSION	No data available
HENRY'S LAW CONSTANT	No data available
AUTO-IGNITION TEMPERATURE	not considered highly flammable
DECOMPOSITION TEMPERATURE	390°C ± 5 °C
VISCOSITY	not considered highly flammable
EXPLOSIVE PROPERTIES	No Explosive
OXIDIZING PROPERTIES	No Oxidising properties

10. STABILITY AND REACTIVITY

CHEMICAL STABILITY:	Stable under recommended storage conditions.
STORAGE STABILITY:	Stable under normal storage conditions in original container for at least two years.
COMPATIBILITY:	Strong oxidizing agents
CONDITIONS TO AVOID	No data available
HAZARDOUS DECOMPOSITION PRODUCTS:	Other decomposition products - No data available

11. TOXICOLOGICAL INFORMATION

ACUTE ORAL LD50	LD50 Oral - Rat 300 - 2000 mg/kg
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ACUTE DERMAL LD50	LD50 Dermal - Rat - > 2,000 mg/kg bw
INHALATION LC50	LC50 Inhalation- Rat > 2.78 mg/l , 4 h
SKIN IRRITATION	Not irritating
EYES IRRITATION	Not irritating
SKIN SENSITISATION	Not a skin sensitizer
SHORT TERM TOXICITY	Relevant oral NOAEL: 1-year, dog: 0.5 mg/kg bw per day 90-day, rat: 2.6 mg/kg bw per day Relevant dermal NOAEL: 21-day, rat: 10 mg/kg bw per day Relevant inhalation NOAEL: 21-day, rat: 0.3 µg/L air (corresponding to an inhaled dose of 0.08 mg/kg bw per day)
DEVELOPMENTAL	Rat : 15 mg/kg bw per day –the highest dose tested
GENOTOXICITY	No genotoxic potential
REPRODUCTIVE/DEVELOPMENTAL EFFECTS:	5.2 mg/kg bw per day
CHRONIC/SUBCHRONIC TOXICITY STUDIES:	1.7 mg/kg bw per day (2-year, rat) 1.8 mg/kg bw per day (2-year, mouse)
CARCINOGENICITY:	No carcinogenic potential

12. ECOLOGICAL INFORMATION

TOXICITY

Acute toxicity to birds:

LD50 > 3950 mg/kg bw (Mallard duck)

Dietary toxicity to birds:

LC50 > 530 mg/kg bw per day (Bobwhite quail)

Acute toxicity fish:

96h-LC50 = 0.24 mg/L (O. mykiss)

Long term toxicity fish:

28 d-NOEC = 0.25 mg/L (Cyprinodon variegatus)

Honeybees:

Acute oral toxicity: LD50 > 0.91 µg/bee

Acute contact toxicity: LD50 > 0.038 µg/bee

PERSISTENCE AND DEGRADABILITY

No data available

BIOACCUMULATIVE POTENTIAL

No data available

13. DISPOSAL CONSIDERATIONS

WASTE TREATMENT METHODS

Product

Offer surplus and non-recyclable solutions to a licensed disposal company. Dissolve or mix the material with a combustible solvent and burn in a chemical incinerator equipped with an afterburner and scrubber.

CONTAMINATED PACKAGING

Dispose of as unused product.

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14. TRANSPORT INFORMATION

IATA

UN-Number 3082

Class 9

PACKING GROUP III

DESCRIPTION OF THE GOODS TOXIC SOLID, ORGANIC, N.O.S. (Lambda-cyhalothrin)

MARINE POLLUTANT: yes

15. REGULATORY INFORMATION

SAFETY, HEALTH AND ENVIRONMENTAL REGULATIONS/LEGISLATION SPECIFIC FOR THE SUBSTANCE OR MIXTURE

This safety datasheet complies with the requirements of Regulation (EC) No. 1907/2006.

16. OTHER INFORMATION

The data given here is based on current knowledge and experience. The purpose of this safety data sheet is to describe the products in terms of their safety requirements. The data does not signify any warranty with regard to the products' properties.